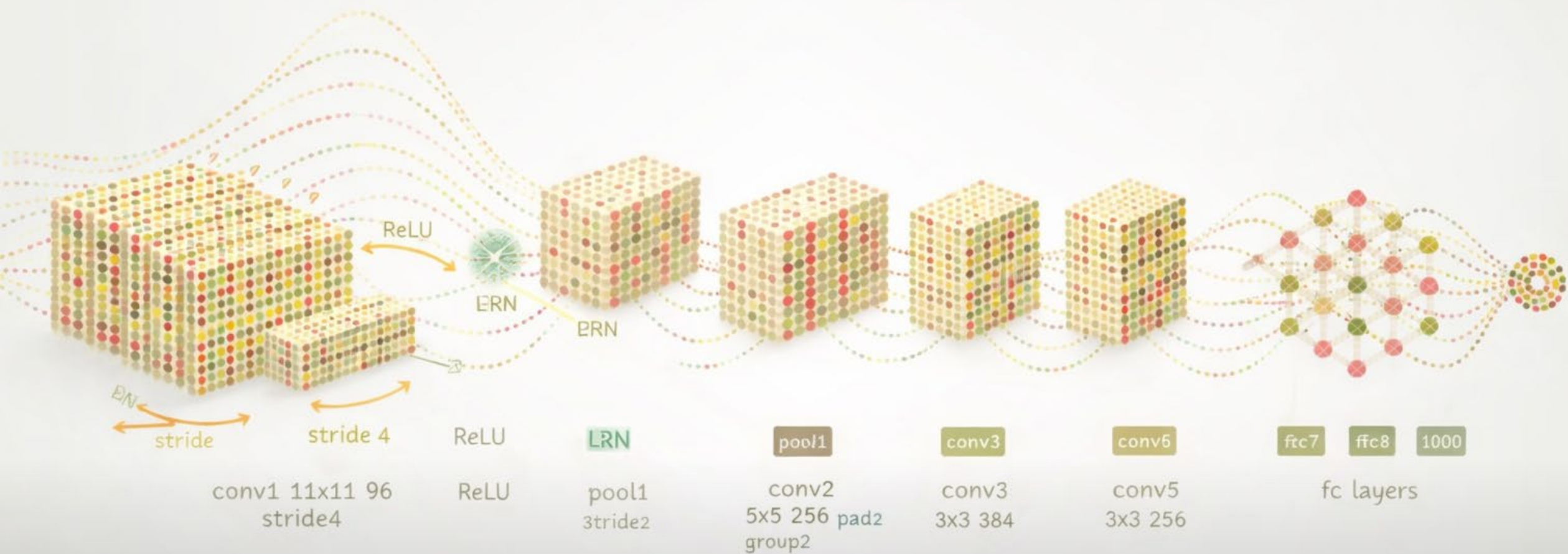




CONVOLUTIONAL NEURAL NETWORKS

TERMS

- ANN = Artificial neural network
- DNN = Deep neural network
- CNN = Convolutional neural network
- RNN = Recurrent neural network

CONVOLUTIONAL NEURAL NETWORKS

- CNNs mimic the structure and function of the visual system
- They use layers of “convolutions” (or filters) to learn and extract features of the input data
- Specifically designed to process and analyze data with a grid-like structure
 - Images
 - Videos

WHAT IS A CONVOLUTION?

- Mathematical operation on two functions (f and g) that produces a third function ($f * g$) that expresses how the shape of one is modified by the other

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t - \tau)d\tau$$

- Convolution is fancy multiplication
 - Convolutions are often described as multiplication and addition over a flipped sliding window
-

CONTINUOUS EXAMPLE

- **Define two simple functions:**

$$f(t) = \begin{cases} 1 & 0 \leq t \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad g(t) = \begin{cases} 1 & 0 \leq t \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- **Convolution definition:**

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau$$

- τ :integration variable
 - t : output location
 - $g(t - \tau)$:flipped and shifted version of g
-



CONTINUOUS EXAMPLE

- $f(\tau)$ is nonzero on $[0, 1]$
- $g(t - \tau)$ is nonzero when $t - 1 \leq \tau \leq t$
- **Therefore:**

$(f * g)(t) = \text{Length of Overlap between the intervals } [0, 1] \text{ and } [t - 1, t]$

DISCRETE EXAMPLE

- Suppose patients come to our hospital this week with a certain disease and a different number of patients come each day
 - M: 1 patients
T: 2 patients
W: 3 patients
Th: 4 patients
F: 5 patient
 - In vector format: [1, 2, 3, 4, 5]
 - Suppose that we treat each patient with 3 doses of medication
 - How many doses of medication will we need each day?
 - [3, 6, 9, 12, 15]
-

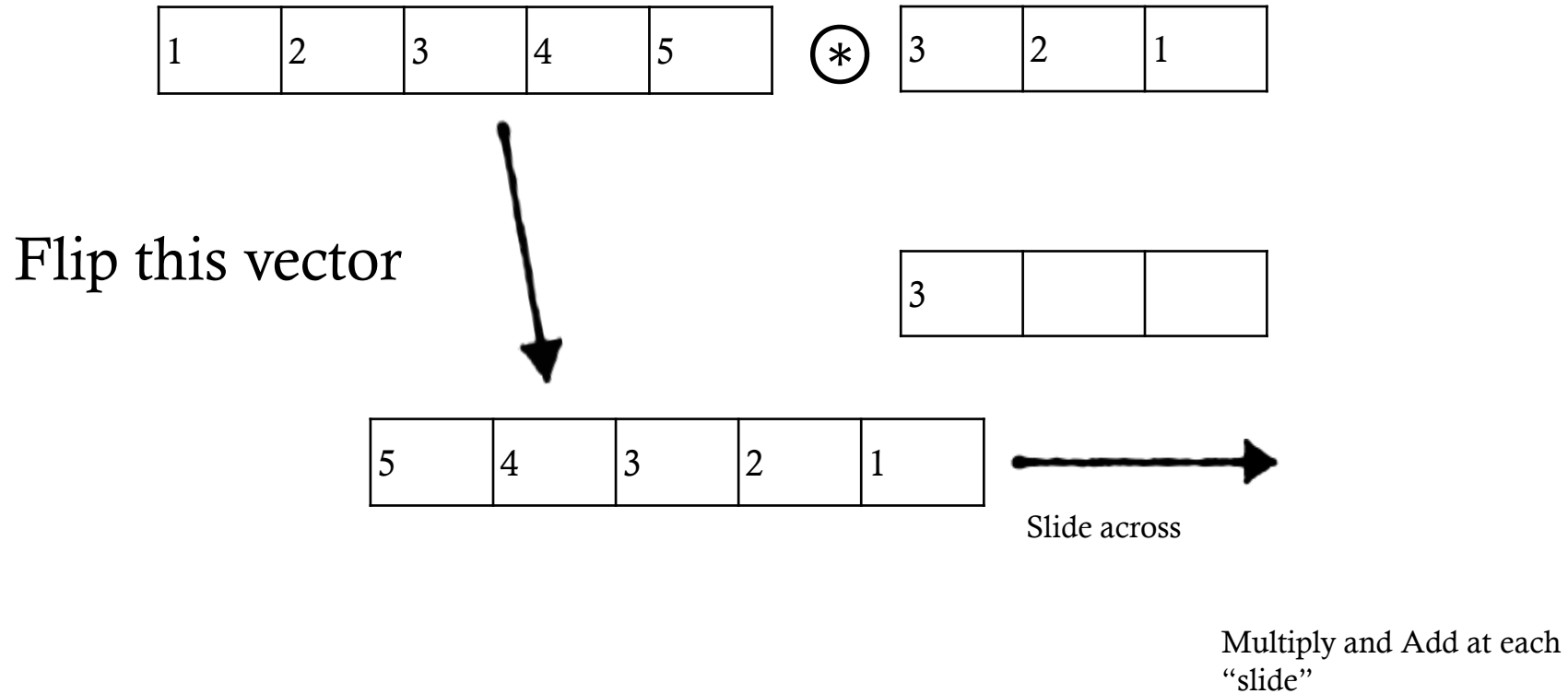
DISCRETE EXAMPLE

- Suppose patients come to our hospital this week with a certain disease and a different number of patients come each day
 - In vector format: $[1, 2, 3, 4, 5]$
 - Now suppose that patients need a multi-day regimen with 3 doses the first day, 2 the second day, and 3 doses the third day
 - $[3, 2, 1]$
 - How many doses of medication will we need each day?
-

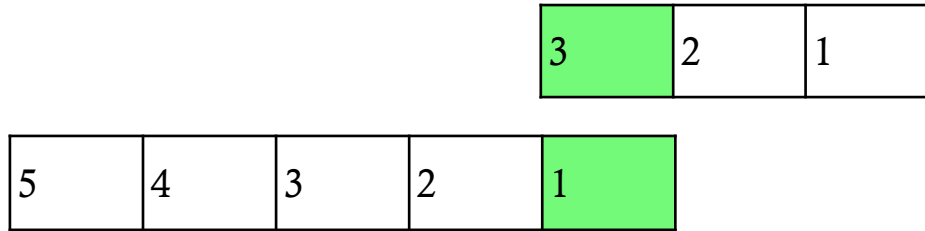
DISCRETE EXAMPLE

- Suppose patients come to our hospital this week with a certain disease and a different number of patients come each day
 - In vector format: [1, 2, 3, 4, 5]
 - Now suppose that patients actually need a multi-day regimen with 3 doses the first day, 2 the second day, and 3 doses the third day
 - [3, 2, 1]
 - How many doses of medication will we need each day?
 - Monday: $1*3$
 - Tuesday: $2*3 + 1*2$
 - Wednesday: $3*3 + 2*2 + 1*1$
 - Thursday: $4*3 + 3*2 + 2*1$
 - Friday: $5*3 + 4*2 + 3*1$
 - Saturday: $5*2 + 4*1$
 - Sunday: $5*1$
-

DISCRETE EXAMPLE



DISCRETE EXAMPLE



$$\begin{bmatrix} 3 & & \end{bmatrix} = \begin{bmatrix} 3 \end{bmatrix}$$

DISCRETE EXAMPLE

3	2	1
---	---	---

5	4	3	2	1
---	---	---	---	---

6	2	
---	---	--

 $=$

8

DISCRETE EXAMPLE

		3	2	1
5	4	3	2	1

9	4	1
---	---	---

 $=$

14

DISCRETE EXAMPLE

	3	2	1	
5	4	3	2	1

12	6	2
----	---	---

 $=$

20

DISCRETE EXAMPLE

3	2	1
---	---	---

5	4	3	2	1
---	---	---	---	---

15	8	3
----	---	---

 =

26

DISCRETE EXAMPLE

3	2	1
---	---	---

5	4	3	2	1
---	---	---	---	---

	10	4
--	----	---

 $=$

14

DISCRETE EXAMPLE

3	2	1
---	---	---

5	4	3	2	1
---	---	---	---	---

		5
--	--	---

 =

5

DISCRETE EXAMPLE

1	2	3	4	5
---	---	---	---	---

 $\circledast$

3	2	1
---	---	---

 =

3	8	14	20	26	14	5
---	---	----	----	----	----	---

```
np.convolve([1,2,3,4,5],[3,2,1])  
array([ 3,  8, 14, 20, 26, 14,  5])
```

USES OF CONVOLUTIONS

- Convolutions have application in a vast array of disciplines
 - Probability
 - Statistics
 - Image processing
 - Signal processing
 - Acoustics
 - Computer vision
 - Etc.
-

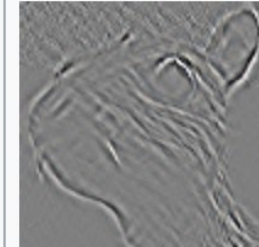
CONVOLUTIONS IN IMAGE PROCESSING

- In image processing, convolving a filter with the original image results in a range of effects, depending on the filter



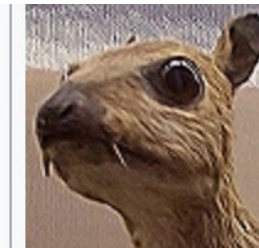
Edge detection

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$



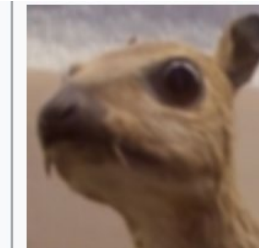
Sharpen

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$



Blur

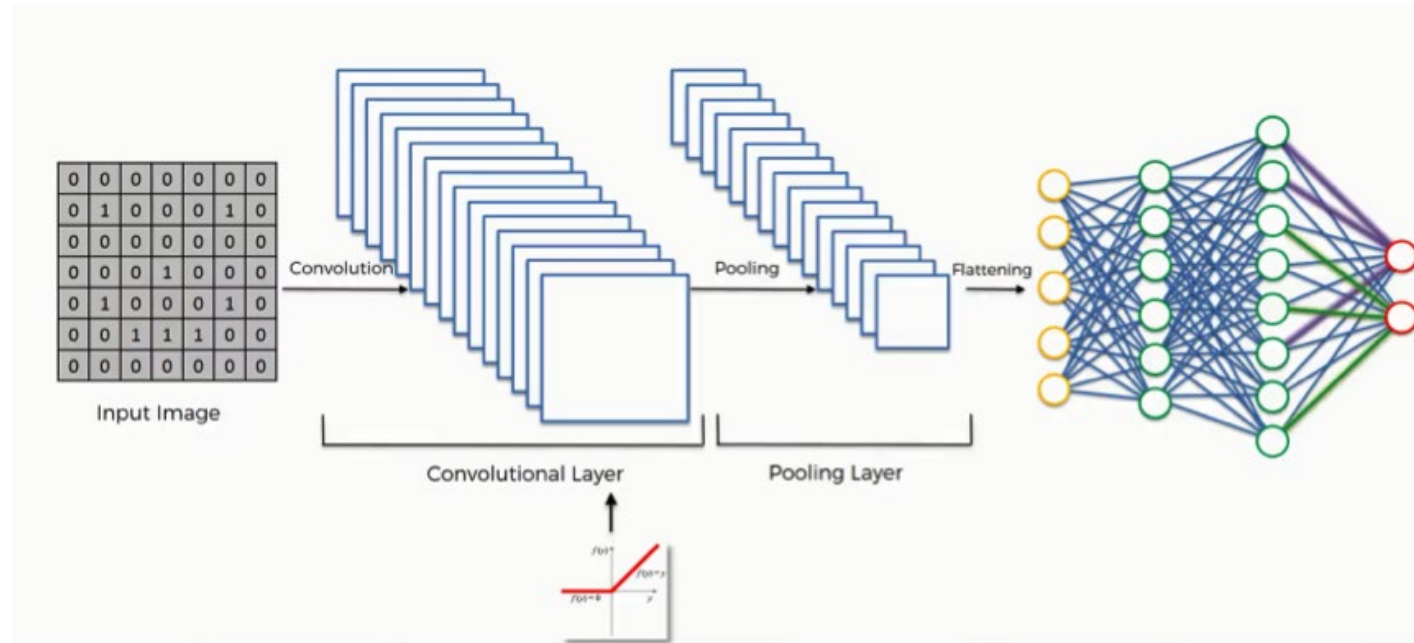
$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$



CNN ARCHITECTURE

- Basic structure of a CNN:
 - Convolutional layers
 - part of fitting process involves the network finding the filters that help identify the image
 - Pooling layers
 - Flattening and Fully connected layers

CNN ARCHITECTURE



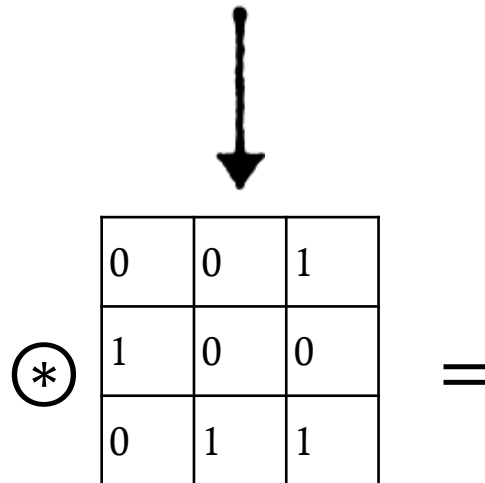
CONVOLUTIONAL LAYER

CONVOLUTION WITH ONE FILTER

This is learned by the model

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image



Feature Detector/
Filter

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0				

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0	0			

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0	0	1		

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0	0	1	0	

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

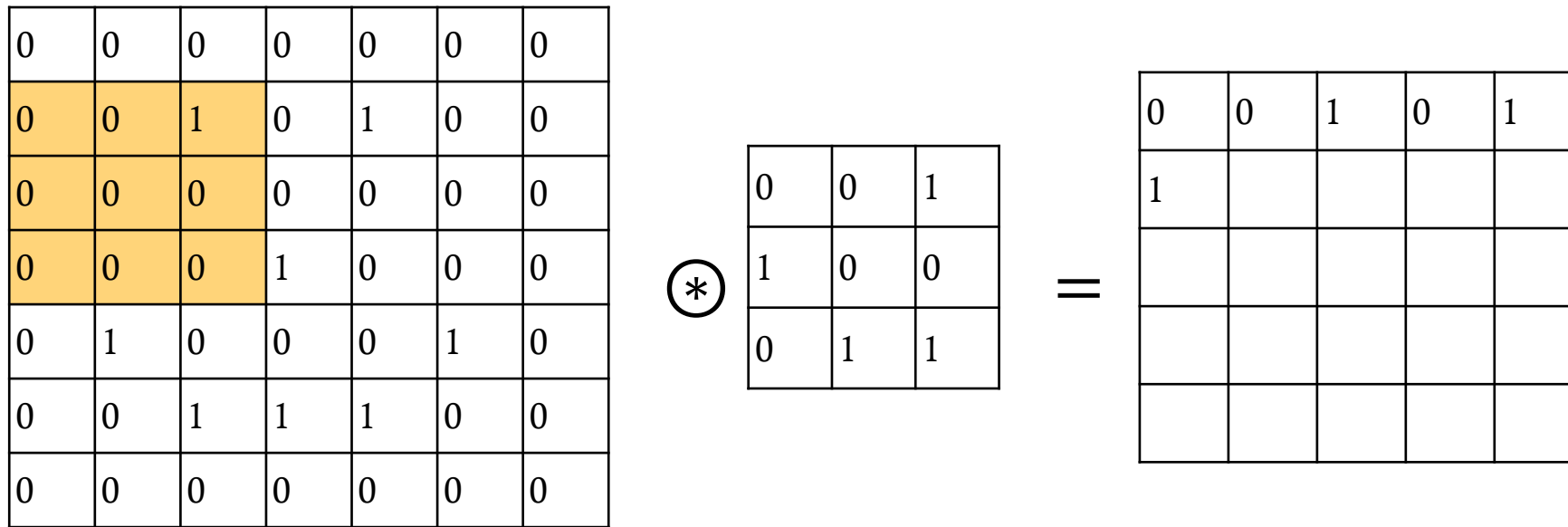
Feature Detector/
Filter

=

0	0	1	0	1

Feature Map

CONVOLUTION WITH ONE FILTER



Input Image

Feature Detector/
Filter

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0	0	1	0	1
1	1			

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0	0	1	0	1
1	1	2		

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0	0	1	0	1
1	1	2	0	

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0	0	1	0	1
1	1	2	0	0

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0	0	1	0	1
1	1	2	0	0
1				

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0	0	1	0	1
1	1	2	0	0
1	0			

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0	0	1	0	1
1	1	2	0	0
1	0	0		

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0	0	1	0	1
1	1	2	0	0
1	0	0	2	

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0	0	1	0	1
1	1	2	0	0
1	0	0	2	1

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0	0	1	0	1
1	1	2	0	0
1	0	0	2	1
1				

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0	0	1	0	1
1	1	2	0	0
1	0	0	2	1
1	4			

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0	0	1	0	1
1	1	2	0	0
1	0	0	2	1
1	4	2		

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0	0	1	0	1
1	1	2	0	0
1	0	0	2	1
1	4	2	1	

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0	0	1	0	1
1	1	2	0	0
1	0	0	2	1
1	4	2	1	0

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0	0	1	0	1
1	1	2	0	0
1	0	0	2	1
1	4	2	1	0
0				

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0	0	1	0	1
1	1	2	0	0
1	0	0	2	1
1	4	2	1	0
0	0			

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0	0	1	0	1
1	1	2	0	0
1	0	0	2	1
1	4	2	1	0
0	0	1		

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

Feature Detector/
Filter

=

0	0	1	0	1
1	1	2	0	0
1	0	0	2	1
1	4	2	1	0
0	0	1	2	

Feature Map

CONVOLUTION WITH ONE FILTER

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

Input Image

$\otimes$

0	0	1
1	0	0
0	1	1

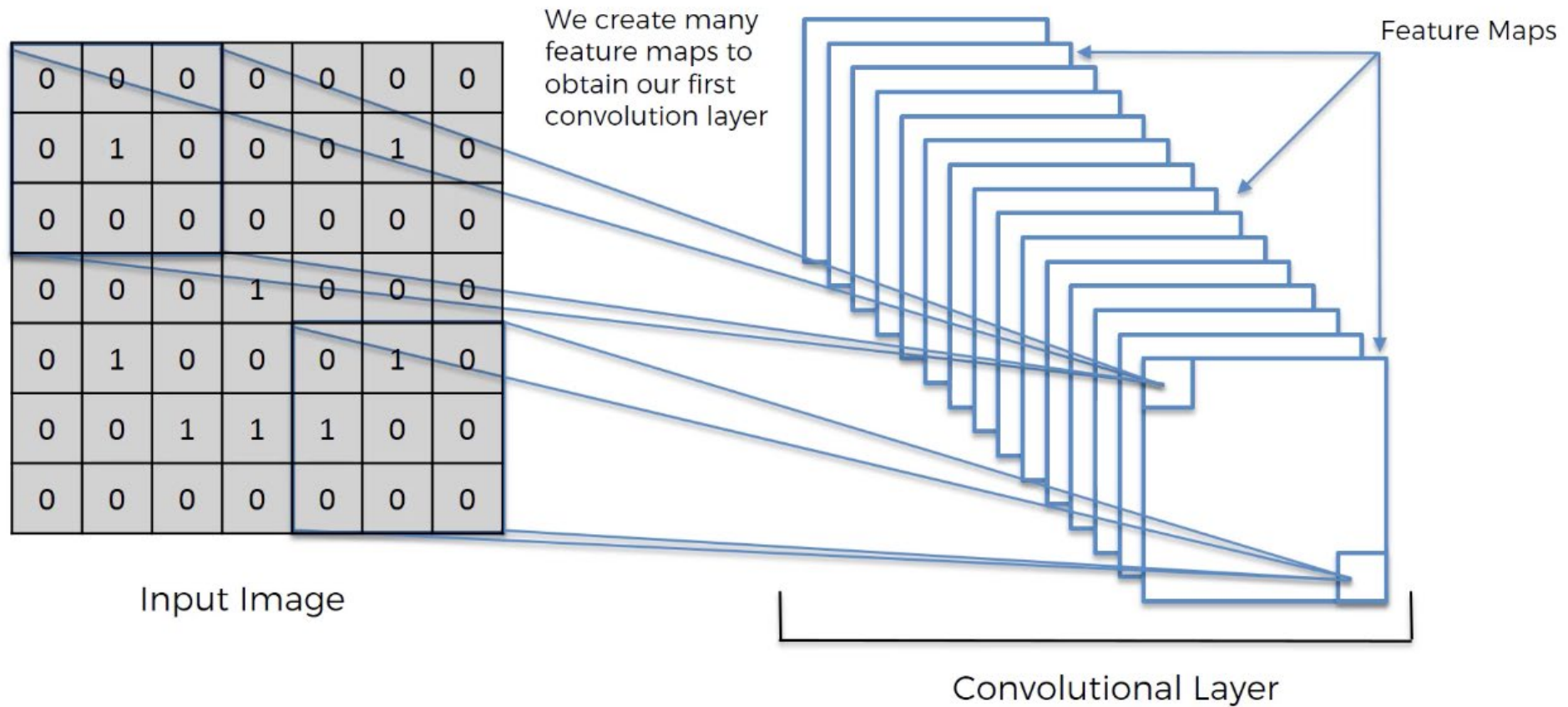
Feature Detector/
Filter

=

0	0	1	0	1
1	1	2	0	0
1	0	0	2	1
1	4	2	1	0
0	0	1	2	1

Feature Map

CONVOLUTIONAL LAYER



POOLING LAYER

MAX POOLING

- Pooling layers operate on the feature maps to summarize the pixels in a rectangular neighborhood
- Max pooling is the most common, though other pooling operations are occasionally used
- The purpose of the pooling layer is to:
 - Reduce the the spatial size of the feature map
 - Help make the network robust to small translations and distortions of the input data

MAX POOLING

0	0	1	0	1
1	1	2	0	0
1	0	0	2	1
1	4	2	1	0
0	0	1	2	1

Feature Map

Max Pooling
→
Box size: 2x2
Stride = 2

1		

Pooled Feature Map

MAX POOLING

0	0	1	0	1
1	1	2	0	0
1	0	0	2	1
1	4	2	1	0
0	0	1	2	1

Feature Map

Max Pooling
→
Box size: 2x2
Stride = 2

1	2	

Pooled Feature Map

MAX POOLING

0	0	1	0	1
1	1	2	0	0
1	0	0	2	1
1	4	2	1	0
0	0	1	2	1

Feature Map

Max Pooling
→
Box size: 2x2
Stride = 2

1	2	1

Pooled Feature Map

MAX POOLING

0	0	1	0	1
1	1	2	0	0
1	0	0	2	1
1	4	2	1	0
0	0	1	2	1

Feature Map

Max Pooling
→
Box size: 2x2
Stride = 2

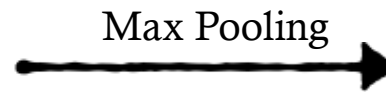
1	2	1
4		

Pooled Feature Map

MAX POOLING

0	0	1	0	1
1	1	2	0	0
1	0	0	2	1
1	4	2	1	0
0	0	1	2	1

Feature Map



Box size: 2x2
Stride = 2

1	2	1
4	2	

Pooled Feature Map

MAX POOLING

0	0	1	0	1
1	1	2	0	0
1	0	0	2	1
1	4	2	1	0
0	0	1	2	1

Feature Map

Max Pooling
Box size: 2x2
Stride = 2



1	2	1
4	2	1

Pooled Feature Map

MAX POOLING

0	0	1	0	1
1	1	2	0	0
1	0	0	2	1
1	4	2	1	0
0	0	1	2	1

Feature Map

Max Pooling
→
Box size: 2x2
Stride = 2

1	2	1
4	2	1
0		

Pooled Feature Map

MAX POOLING

0	0	1	0	1
1	1	2	0	0
1	0	0	2	1
1	4	2	1	0
0	0	1	2	1

Feature Map

Max Pooling
→
Box size: 2x2
Stride = 2

1	2	1
4	2	1
0	2	

Pooled Feature Map

MAX POOLING

0	0	1	0	1
1	1	2	0	0
1	0	0	2	1
1	4	2	1	0
0	0	1	2	1

Feature Map

Max Pooling
→
Box size: 2x2
Stride = 2

1	2	1
4	2	1
0	2	1

Pooled Feature Map

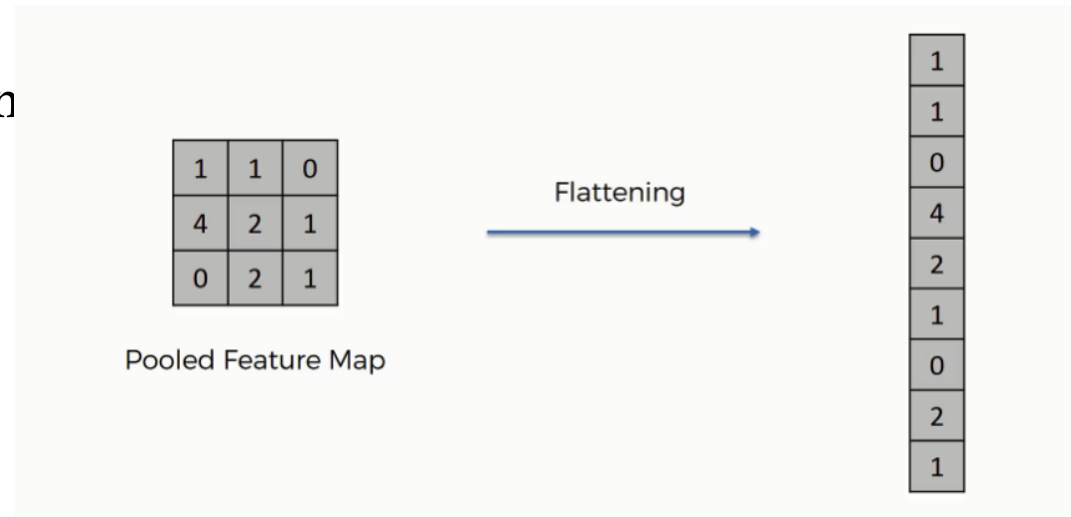
FLATTENING & FULLY CONNECTED LAYERS

IMAGE TO CLASSIFICATION

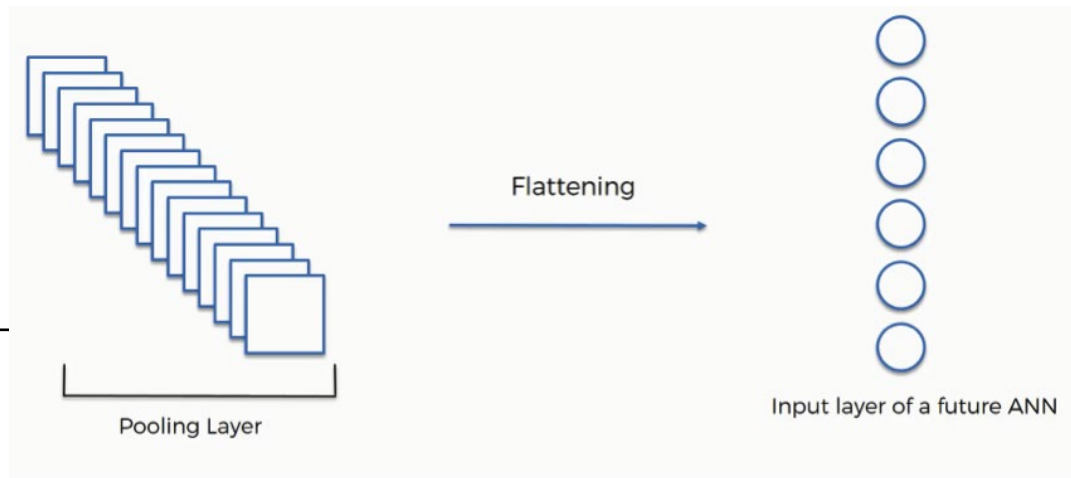
- After one or more convolution and pooling layers
- Send the result into a regular ANN
 - Combine features into a wider variety of attributes
 - Perform the classification step

FLATTENING

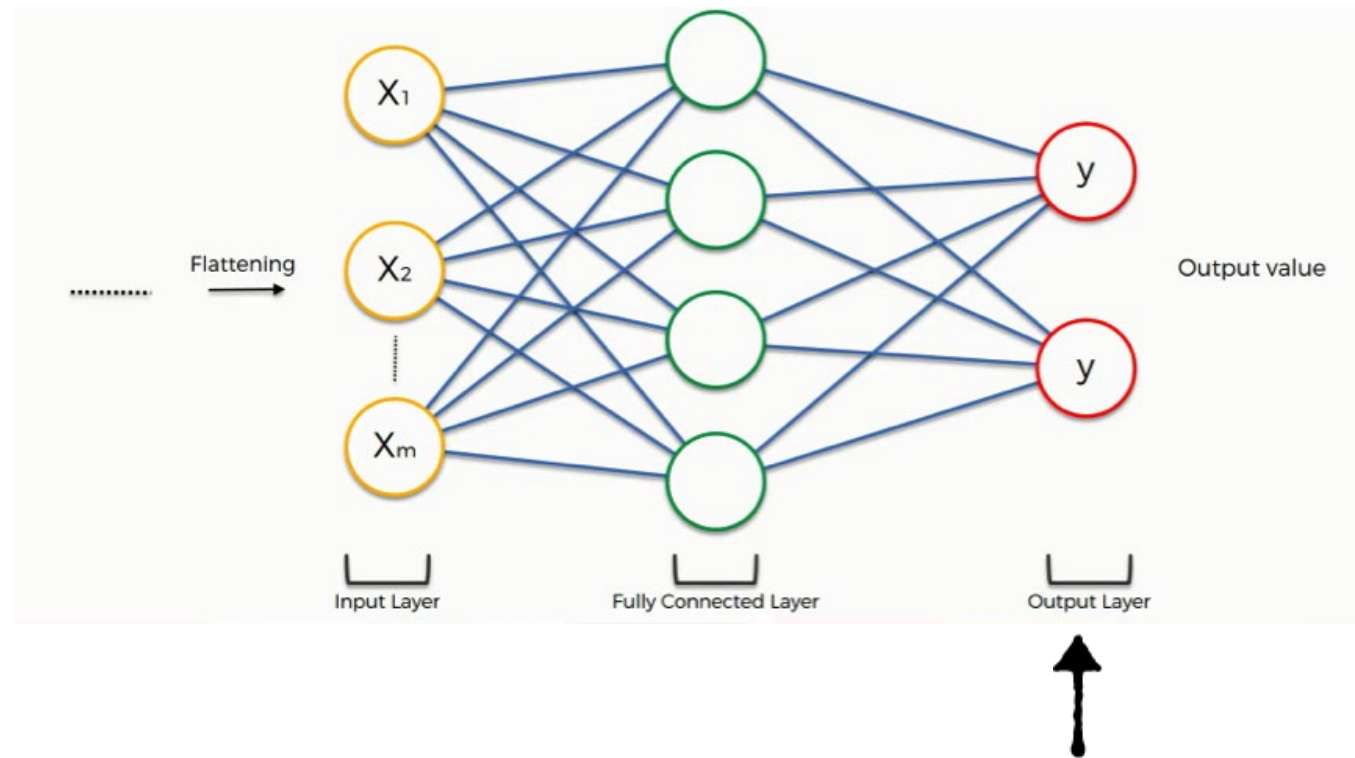
- Flattening simply mean
ANN



input into a regular

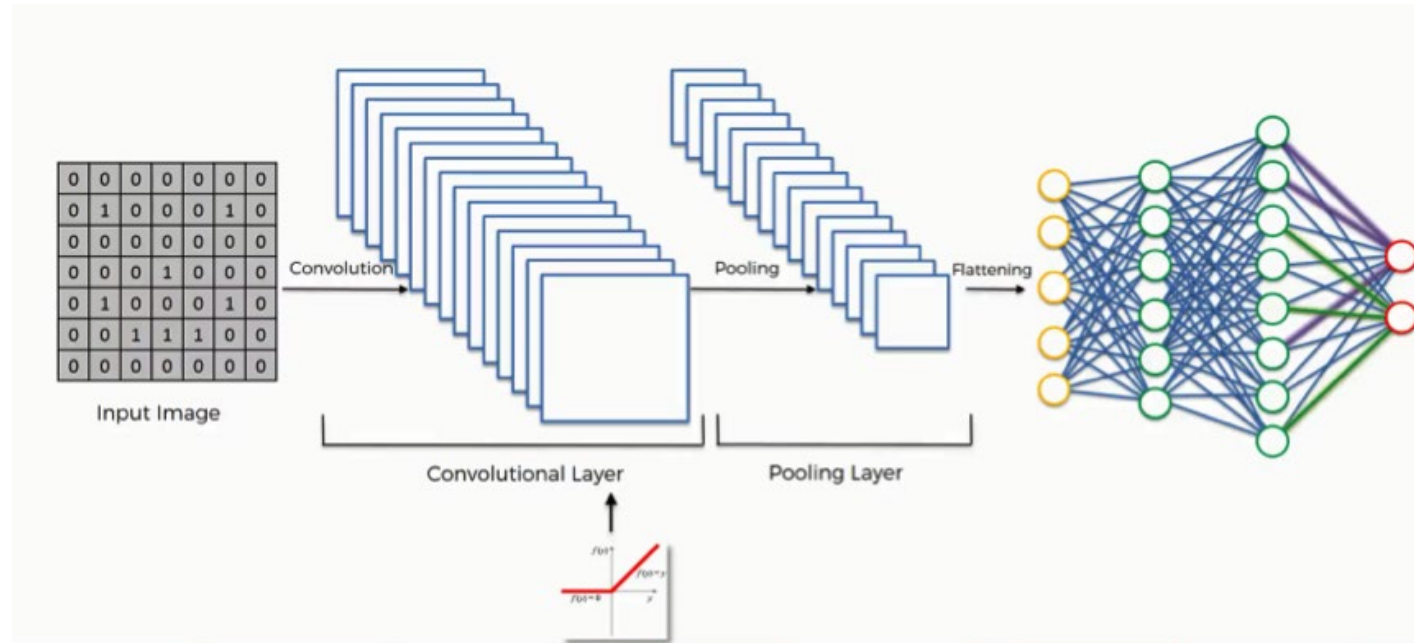


FULLY CONNECTED LAYER



The output layer will have a neuron for every possible class

SUMMARY

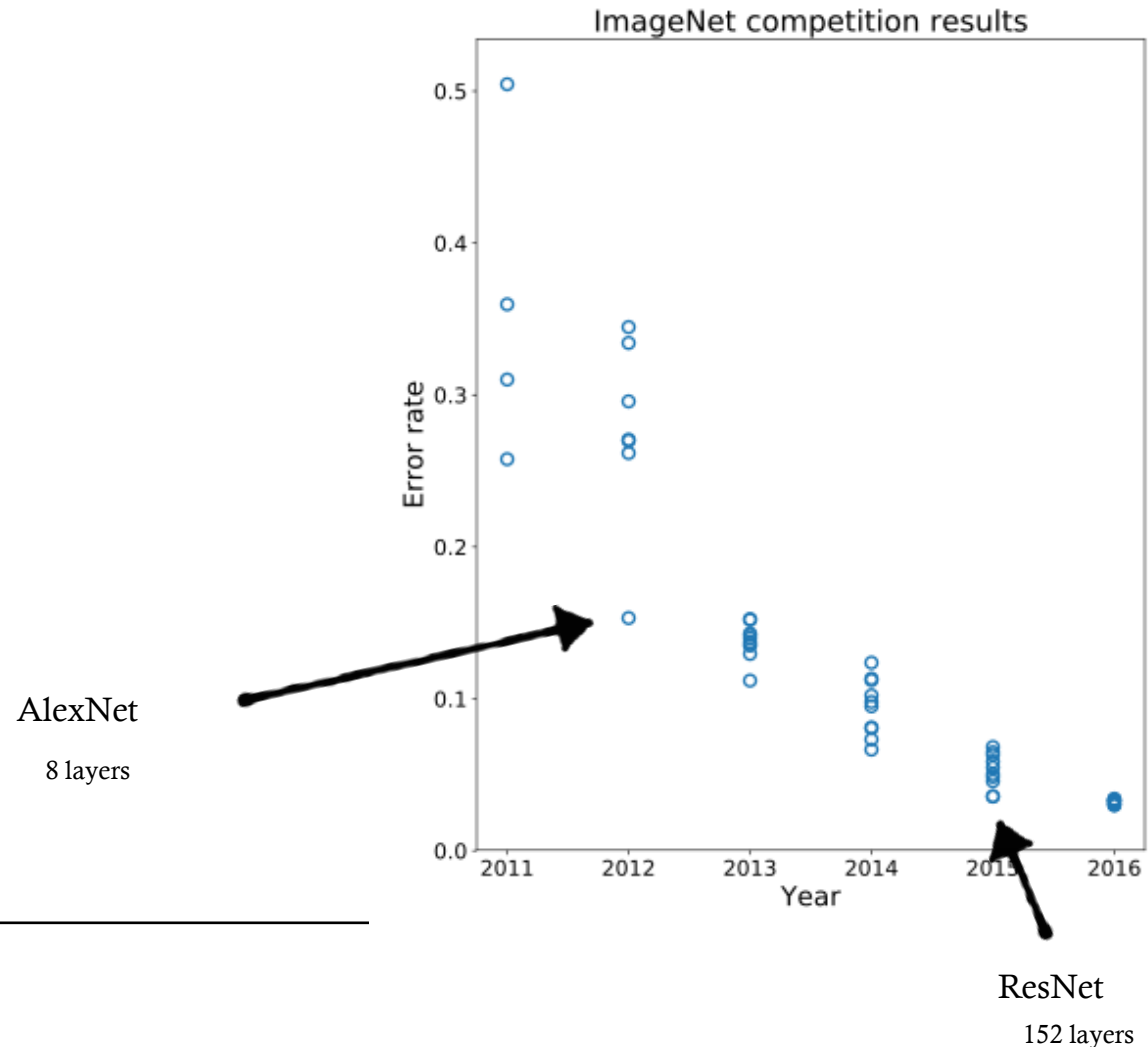


IMAGENET

- Large visual database designed for object recognition research
 - ~14 million images
 - Hand-annotated (Mechanical Turk) into 20,000+ categories
 - Subset of the data is used in the ImageNet Large Scale Visualization Recognition Challenge (ILSVRC)
 - Help end AI winter
 - In 2012 AlexNet achieved a top-5 error rate more than 10.8 percentage points lower than the runner up
 - “Suddenly people started to pay attention, not just within the AI community but across the industry as a whole”
(from *The Economist*)
-

IMAGENET LARGE-SCALE VISUAL RECOGNITION CHALLENGE

- Subset of ImageNet Data
- 1.2 million training images
- 1000 classes



MODERN CNN ARCHITECTURES

VARIOUS CNN ARCHITECTURES

- We should think of the word “architecture” fairly literally
- It takes time, education, and practice to become an “architect”
 - True for architects that design buildings
 - True for architects that design CNNs
- We can learn from what others have done and other architectures that have been successful

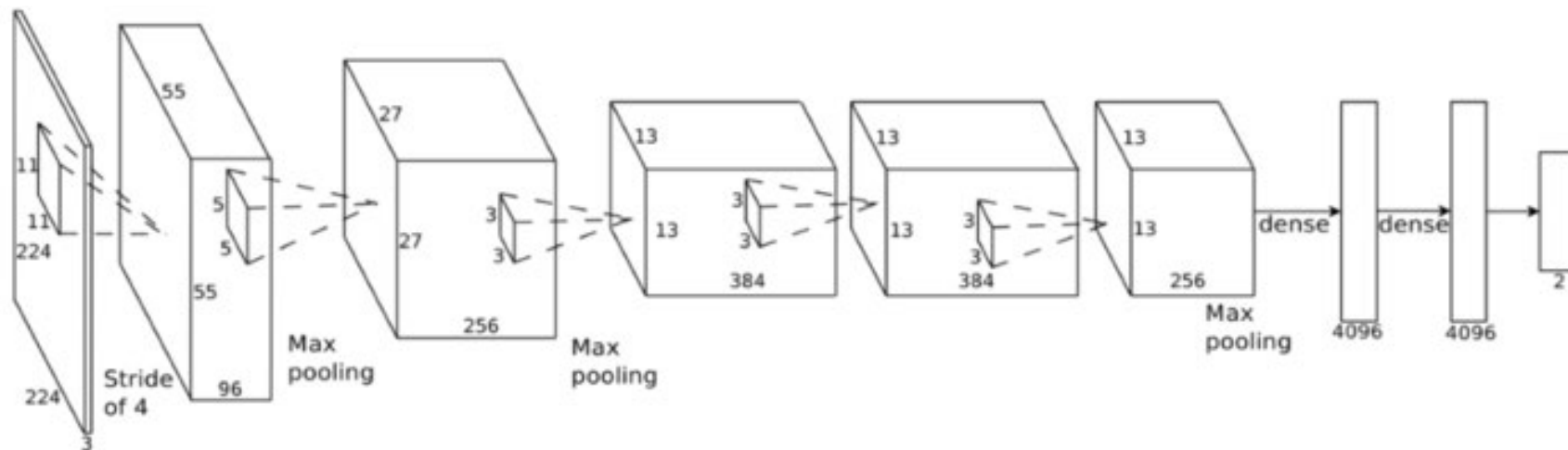


INNOVATIONS THAT MADE DEPTH POSSIBLE

- ReLU (and its relatives)
- Skip connections
- Better weight initializations
- Batch normalization
 - Normalize the inputs of each layer (helps with internal covariate shift, vanishing gradients, overfitting)
- Dropout
 - Randomly dropout or “turn off” some of the neurons during training
 - Forces the network to have redundant representation of features
- (Not an exhaustive list)

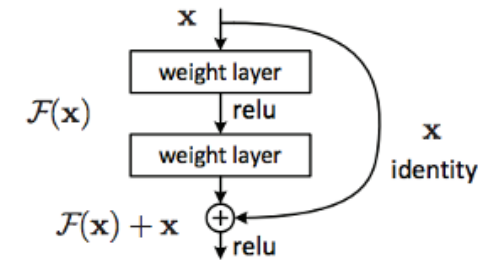
ALEXNET

- Innovations
 - Deep convolutions
 - ReLu
- Also:
 - Dropout
 - Data augmentation
 - GPU



RESNET (RESIDUAL NETWORK)

- Introduced by researchers at Microsoft in 2015
- ResNet can train much deeper networks while avoiding the vanishing gradient problem
- Innovation:
 - Skip connection: allow the gradient to pass thru fewer steps
 - Residual block: allow network to learn residual functions
 - Enables the network to “skip” some layers and focus on learning residual functions that cannot be learned by the shallower layers
- One of the most widely used deep learning architectures used today



CNN TRAINING

TRAINING COMPONENTS

- Initialization
 - Forward pass
 - Compute loss
 - Backpropagation
 - Update weights
 - Repeat over mini-batches
 - Key architectural hyperparameters:
 - Kernel (filter) size
 - Stride
 - Padding
 - Number of filters
-

CNN LAYER COMPONENTS

A convolutional layer is defined by:

- **Kernel size ($K \times K$)** → sliding window size
- **Stride (S)** → step size of movement
- **Padding (P)** → zeros added around border
- **Number of filters (F)** → number of feature maps
- Output spatial size:

$$\text{Output size} = \frac{N - K + 2P}{S} + 1$$

KERNEL (SLIDING WINDOW)

What does kernel size control?

- Spatial pattern scale detected
- Parameter count
- Parameters per filter:

$$K \times K \times C$$

- Where C is the number of input channels (typically 1 or 3).
-

KERNEL SIZE

- Larger kernels:
 - More parameters
 - Higher variance
 - More compute
 - Larger receptive field per layer
 - Smaller kernels (e.g., 3×3):
 - Fewer parameters
 - Deeper networks needed
 - Better regularization
 - More nonlinearities
 - This is why modern CNNs prefer stacked 3×3 convolutions.
-

STRIDE

- Stride = how far the filter moves each step.
 - $S = 1 \rightarrow$ dense coverage
 - $S = 2 \rightarrow$ downsampling
- Effect on output size:

$$\frac{N - K + 2P}{S} + 1$$

- Larger stride:
 - Smaller output feature maps
 - Fewer activations
 - Fewer gradient paths
 - Faster training
 - More aggressive information compression
 - Smaller stride:
 - More overlap
 - More gradient signal
 - Better fine-detail learning
 - Higher computation cost
-

PADDING

Padding adds zeros around input borders.

- Types:
 - No padding
 - Same padding (preserve size)
 - Without padding:
 - Spatial size shrinks each layer
 - With padding:
 - Spatial resolution preserved
- Without padding:
 - Border pixels used fewer times
 - Gradients weaker at edges
 - Spatial size shrinks quickly
 - With padding:
 - More uniform gradient flow
 - Deeper networks possible
 - Preserves spatial alignment
 - Padding helps stabilize training in deep CNNs.
-

WEIGHT SHARING

Fully connected (no sharing)

- If you flatten a 224×224 image:
 - Input size $\approx 50,000$ pixels
 - If next layer has 100 neurons
 - You need $50,000 \times 100 = 5,000,000$ **weights**
 - Each neuron has its own independent weights.
- Suppose:
 - Kernel size = 3×3
 - Input has 3 channels (RGB)
 - Number of filters = 64
 - Each filter has:
 - $3 \times 3 \times 3 = 27$ *weights*
 - Total parameters:
 - $27 \times 64 = 1728$ *weights*
 - Those 27 weights for a filter are used at **every spatial location**
 - We do NOT learn different weights for different pixels
-

USING PRE-TRAINED MODELS

PRE-TRAINED MODELS

- Many of these very large neural networks are publicly available
- These models can be used:
 - In Transfer Learning
 - Use the lower layers of an existing DNN
 - Train the upper layers according to the specific new task
 - Directly
 - If you have a similar task and similar outputs